OOP Assignment 3 - Library Management System

Project Overview

This project is a Java-based Object-Oriented Library Management System. It consists of four main classes: Author, Book, Library, and Main. The primary technical focus of this assignment is managing bidirectional object relationships, utilizing constructor chaining, and manually handling data within fixed-size standard arrays without relying on dynamic collections like ArrayList.

File Structure

- **Author.java**: Represents an author, holding a fixed-size array of books they have written.
- **Book.java**: Represents a book, holding a fixed-size array of its authors and formatting methods.
- **Library.java**: The central management class that holds an inventory of books and processes search, add, and remove operations.
- **Main.java**: The testing ground to instantiate objects and verify all required functionalities.

Implementation Details

1. The Author Class

- **Constructor Chaining:** The assignment required specific chaining logic. The first constructor initializes the booksWritten array with a default capacity of 10. The second constructor uses this(authorId, name) on its first line to satisfy the chaining requirement, and then safely re-initializes the array using the custom maxBooks parameter.
- **Data Management:** The addBook method checks if the internal bookCount is less than the array's capacity before adding a new book, preventing ArrayIndexOutOfBoundsException errors.

2. The Book Class

- **Bidirectional Relationship:** The most critical logic resides in the addAuthor(Author author) method. When an author is added to a book's authors array, the method

explicitly calls author.addBook(this). This guarantees that both objects remain synchronized without needing to add them to each other manually in the Main class.

- **Data Validation:** The setPrice(double price) setter includes a validation check. If a negative value is passed, it is automatically corrected to 0.0.
- **String Formatting:** The getReferenceAPA() method formats the primary author's name, the publication year, the book title, and the publisher into a standard APA 7th edition citation string.

3. The Library Class

- **Array Shifting (Removal Logic):** Because the project relies on standard arrays, removing a book in removeBook(int bookId) required manual memory management. The method first locates the index of the target book. Then, it uses a loop starting from that index to shift all subsequent books one position to the left (inventory[i] = inventory[i + 1]). Finally, it nullifies the last duplicated element and decrements the inventoryCount.
- **Nested Searching:** The searchBookByAuthor(String authorName) method uses nested loops. It iterates through the library's inventory, and for each book, it iterates through that book's authors array. If a match is found, it triggers the book's display method.
- **Safe Math:** The getAveragePrice() method sums up the prices of all active books in the inventory. It includes a safeguard to immediately return 0.0 if the library is empty, preventing a divide-by-zero error.

How to Run

1. Ensure all four .java files are in the same directory/package (Assignment03).
2. Compile the files**: javac Assignment03/*.java**
3. Run the main application: **java Assignment03.Main**